Green University of Bangladesh

Department of Computer Science and Engineering (CSE)
Semester: (Spring, Year: 2026), B.Sc. in CSE (Day)

InsightCart

Course Title : Machine Learning Lab
Course Code : CSE 404
Section : 231-D2

Students Details

Name	ID
Promod Chandra Das	231002005
Md. Shahed Shamim Rony	231902019

Submission Date : 28-02-2026
Course Teacher's Name : Ms. Mayeesha Farjana

[For teachers use only: Don't write anything inside this box]

Lab Project Status	
Marks:	**Signature:**
Comments:	**Date:**

Contents

Chapter 1

Introduction

1.1 Overview

InsightCart is a Machine Learning based retail analytics project that focuses on analyzing customer transaction data to understand shopping behavior and improve business decision-making. In this project, we plan to use a retail dataset containing customer purchases and apply different Machine Learning and data mining techniques.

The system will analyze transaction records to identify customer segments, frequent product combinations, purchasing trends and unusual transactions. Techniques such as **Clustering, Predictive Model and Anomaly Detection** will be used to extract meaningful insights from the data. The overall objective is to build an intelligent retail analysis system that can support marketing, inventory, and strategic business decisions.

1.2 Motivation

Retail businesses generate a large amount of transaction data every day. However, raw data alone does not provide useful information unless it is properly analyzed. Understanding customer behavior, product demand and purchasing patterns is essential for improving sales and customer satisfaction.

The motivation of this project is to apply Machine Learning concepts learned in our course to a real-world retail dataset. By working on InsightCart, we aim to bridge the gap between theoretical knowledge and practical implementation. This project will also help us strengthen our skills in data preprocessing, model building, evaluation and result interpretation.

1.3 Problem Definition

1.3.1 Problem Statement

Retail stores manage thousands of customer transactions daily. Identifying high-value customers, frequently purchased products and unusual patterns manually is difficult and time-consuming.

There is a need for an intelligent system that can automatically analyze large transaction datasets and generate useful insights. Such a system should be able to segment customers, discover product associations, predict behavior patterns and detect anomalies to support better business decisions.

1.3.2 Complex Engineering Problem

Table 1.1: Summary of Attributes Related to **InsightCart**

Name of the Attribute	**Explanation on How to Address**
P1: Depth of knowledge required	Requires understanding of basic Machine Learning techniques such as clustering, classification, and association rules.
P2: Range of conflicting requirements	Balancing model accuracy with processing time for large datasets.
P3: Depth of analysis required	Analyzing customer behavior and evaluating model performance.
P4: Practical Constraints	Handling large data, missing values, and proper feature selection.

1.4 Design Goals/Objectives

The primary objective of the **InsightCart** project is to design a Machine Learning based system that can analyze retail transaction data and generate intelligent insights for decision support.

- **Customer Segmentation:** Group customers based on purchasing behavior using clustering techniques.
- **Market Basket Analysis:** Identify frequent itemsets and product associations.
- **Predictive Modeling:** Build models to classify customer behavior and predict trends.
- **Anomaly Detection:** Detect unusual or suspicious transactions.
- **Data Visualization:** Present insights through charts and summary reports.
- **Educational Objective:** Apply Machine Learning concepts in a practical retail analytics scenario.

1.5 Application

InsightCart has applications in both academic and real-world retail environments.

- **Retail Marketing:** Helps identify target customers and design personalized promotions.
- **Product Management:** Assists in product bundling and inventory planning.
- **Business Decision Support:** Supports data-driven strategic decisions.
- **Fraud Detection:** Identifies abnormal transaction patterns.
- **Academic Learning:** Provides hands-on experience in applying Machine Learning techniques to real datasets.